Context-Aware RL for IR/RGB Registration and Augmentation in Marine Perception

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***Abstract*—Robust marine perception relies on multi-spectral data (IR/RGB), but aligning and augmenting these distinct modalities under varying sea states is challenging. Static preprocessing pipelines often fail to adapt to rapid environmental changes like glare, fog, or thermal crossover. This proposal introduces a deep reinforcement learning (RL) policy that dynamically selects image registration transforms and augmentation recipes per frame. By conditioning on real-time image quality metrics and environmental context, the agent maximizes downstream detection performance and cross-modal consistency. We benchmark this adaptive approach against static registration, standard AutoAugment, and random policies, evaluating robustness on held-out weather conditions and sensor degradation scenarios.**

I. Introduction

Marine autonomy increasingly depends on paired IR/RGB sensing, but robust perception requires both geometric alignment and domain-robust preprocessing. Fixed registration pipelines and static augmentation schedules often fail under glare, fog, thermal crossover, and platform motion. This is particularly problematic in maritime scenes where frame quality and cross-modal discrepancy vary significantly over time.

This proposal investigates a context-aware RL policy that jointly selects registration strategy and augmentation intensity. Unlike search-then-freeze pipelines such as AutoAugment and RandAugment [1], [2], our policy adapts per frame (or short temporal window) using online quality/state signals. The hypothesis is that dynamic control yields better robustness under environmental shift than static preprocessing.

Recent multimodal detection studies show that explicit discrepancy handling and dynamic fusion materially improve results [3]–[5]. Building on that evidence, we formulate preprocessing as a closed-loop control problem (Fig. **??**) and position our method against prior capabilities in the related-work matrix (Fig. **??**).

II. Approach

We formulate preprocessing control as a Markov Decision Process (MDP) in which an RL agent selects both registration and augmentation actions. As shown in Fig. **??**, each step follows a closed loop:

Input frame pair $\rightarrow$ policy action $\rightarrow$ processed pair $\rightarrow$ detector feedback $\rightarrow$ reward.

A. State and Action Space

State s_t contains frame-quality statistics (e.g., blur, entropy), an initial cross-modal mismatch proxy (e.g., edge or mutual-information consistency), and optional context tags (day/night, fog/clear, rain/clear).

Action a_t has two coupled components:

1) **Registration family selection** $T \in \{\mathrm{Rigid}, \mathrm{Affine}, \mathrm{Homography}\}$.
2) **Augmentation parameters** λ_{aug} controlling magnitude/probability of operations such as blur, thermal noise injection, intensity transforms, and modality-consistent perturbations.

B. Reward and Learning Objective

After action execution, the downstream detector runs on processed pairs and emits task feedback. Reward is

$$r_t = \alpha\, \Delta\mathcal{M}_{task} + \beta\, \Delta\mathcal{M}_{cons} - \gamma\, \mathcal{C}_{comp}, \tag{1}$$

where $\mathcal{M}_{task}$ is detection quality (e.g., mAP), $\mathcal{M}_{cons}$ measures cross-modal consistency, and $\mathcal{C}_{comp}$ penalizes latency.

C. Policy Optimization

We use PPO for stable policy updates in mixed discrete/continuous action spaces. To mitigate sparse reward from expensive detector retraining, we use a lightweight proxy evaluator during early policy learning and periodically refresh with full-detector validation. This setup directly targets the gap left by static augmentation and single-transform pipelines [1], [2], [6], [7].

III. Evaluation

A. Experimental Setup

We evaluate on maritime and multimodal IR/RGB benchmarks with explicit domain-shift slices (e.g., day-to-night, clear-to-fog) and misalignment stress conditions. Splits are organized by condition and sequence to avoid leakage.

B. Baselines

We compare against five classes of methods:

1) **Classical fixed registration**: feature-based matching + RANSAC.
2) **Fixed augmentation**: static hand-designed augmentation schedule.
3) **RandAugment**: global randomized augmentation policy [2].

4) **AutoAugment/PBA**: offline searched static policy [1], [8].
5) **Context-free RL**: same control pipeline without context features.

C. Metrics

Primary metrics are mAP@50 (detection) and mean reprojection error (alignment quality). Secondary metrics include performance drop under induced misalignment, robustness under adverse weather/lighting shifts, and preprocessing latency overhead.

D. Hypothesis Validation

Success is defined as outperforming static augmentation and fixed-registration baselines on the hardest shift splits while maintaining acceptable compute overhead. We also analyze action-selection behavior to verify policy adaptivity (e.g., choosing conservative transforms when feature support is weak). Published evidence motivating this setup is summarized in Fig. **??** [3]–[5].

IV. Conclusion

This proposal targets the instability of static preprocessing in dynamic marine environments. We present an RL-driven framework that treats image registration and augmentation as adaptive decisions rather than fixed hyperparameters. By conditioning on real-time content, the system balances alignment accuracy with data diversity.

We expect to demonstrate that instance-level adaptation is superior to global "search-then-freeze" strategies for multimodal sensor fusion. This work contributes to the broader goal of autonomous systems that self-tune their perception pipelines in response to changing real-world conditions.

References

[1] E. D. Cubuk, B. Zoph, D. Mane, V. Vasudevan, and Q. V. Le, "Autoaugment: Learning augmentation strategies from data," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 113–123.

[2] E. D. Cubuk, B. Zoph, J. Shlens, and Q. V. Le, "Randaugment: Practical automated data augmentation with a reduced search space," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2020, pp. 3008–3017.

[3] R. Vadidar, M. Salehi *et al.*, "A unified rgbt framework for object detection and saliency detection," *arXiv preprint arXiv:2206.03943*, 2022.

[4] Y. Li, X. Li *et al.*, "A rgbt object detection method for maritime environment using adaptive dual-calibration network," *Remote Sensing*, vol. 15, no. 19, p. 4887, 2023.

[5] M. Li, L. Ma, Y. Wang, Y. Fang, J. Liu, and C. Lan, "Fd2-net: Feature difference and dynamic fusion network for infrared and visible object detection," *arXiv preprint arXiv:2412.09258*, 2024.

[6] J. Zhang, C. Wang, S. Liu, L. Lan, and D. Zeng, "Content-aware unsupervised deep homography estimation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 6533–6542.

[7] S. Miao, Z. J. Wang, and R. Liao, "A cnn regression approach for real-time 2d/3d registration," *IEEE Transactions on Medical Imaging*, vol. 35, no. 5, pp. 1352–1363, 2016.

[8] D. Ho, E. Liang, X. Chen, I. Stoica, and P. Abbeel, "Population based augmentation: Efficient learning of augmentation policy schedules," in *International Conference on Machine Learning (ICML)*, 2019, pp. 2731–2741.